Assignment: Pro GAN

Implement a Progressive Growing Generative Adversarial Network (ProGAN) to generate synthetic images of your favourite class from the CIFAR-10 dataset at progressively increasing resolutions, from 32×3232 \times 32 to 128×128128 \times 128 pixels.

Tasks:

- Select your favourite class from the CIFAR-10 dataset and prepare the corresponding training images.
- Implement ProGAN to generate synthetic images belonging to the selected class.
- Train the model by progressively increasing the image resolution from 32×3232 \times 32 to 64×6464 \times 64 and finally to 128×128128 \times 128 pixels.
- Generate and save synthetic images at each resolution stage.
- Evaluate the quality of the generated images through visual inspection and, if feasible, quantitative metrics.
- Document the model architecture, training procedure, experimental results, and challenges encountered.

Submit the source code, generated images at all three resolutions, and a report describing the methodology and results.